Xedni affine-lift optimization: seven- and ten-coefficient paths

Date: 2026-11-01 Updated: 2026-11-02

Outcome

The coefficient-lifting seam of prime-field xedni now has two executable, certifying paths exposed as `ca-icx xedni-lift`. Both use high-precision LLL, **exact-rational** Babai closest-plane selection, and a bounded exact search. The seven-coefficient Weierstrass path ranks by discriminant; the new ten-coefficient general-cubic path implements the linear Step-7 height objective for all `2 <= r <= 9`. Both reject any direction that changes a supplied point-incidence row or coefficient residue class and recheck those properties on the result.

The frozen p=67 comparison has **not run**. The mandatory pre-run checkpoint upload is denied by both the installed presigned token (HTTP 403) and the 2026-11-02 managed AWS identity (`PutObject AccessDenied`). The exact blocker and exit condition are recorded in problems/xedni-affine-cvp-checkpoint-access.md. Consequently this report makes no empirical claim that the new nonlinear search beats the 2000 implementation.

Claim labels used below are **source**, **derived**, **verified control**, **implemented but unmeasured**, and **blocked**.

What “the nine points” actually means

Source. JKSST Section 3.2 chooses an integer `r` with `2 <= r <= 9`, saying that `r=4..6` is most likely. A general projective cubic has ten coefficients up to the incidence equations, so nine independent point rows are the maximum the construction permits; the algorithm does not require a special fixed set of nine points.

For `r>=4` in general cubic form, the paper first sends four points to a fixed projective frame. Steps 6–7 lift the remaining point coordinates and solve an affine coefficient system. Steps 9–10 vary kernel vectors to reduce the discriminant and apply a rank heuristic. The authors’ implementation already used the Havas–Majewski–Matthews HNF algorithm. Therefore replacing a raw integer solve by “some lattice reduction” is not itself a new improvement over their implementation.

Implemented algorithms

Let `u*` be one feasible coefficient vector, let the rows of `D` be admissible coefficient changes, let `B` be the lifted-point incidence matrix, and let `q=Mp` be the required modulus. The input certificate checks

```
B D_i = 0                    and                    D_i = 0 (mod q)
```

for every direction. The optimizer then:

1. LLL-reduces `D` with the repository’s 2048-bit fixed-point Gram–Schmidt implementation at delta 0.99;
2. applies exact-rational Gram–Schmidt and Babai nearest-plane to target `-u*`, retaining arbitrary-precision integer center coefficients;
3. enumerates a bounded neighborhood around the Babai coefficients;
4. uses the model-specific exact objective; and
5. certifies `B u = B u*` and `u = u* (mod q)` on the selected vector.

For the Weierstrass model, step 4 rejects singular curves and orders candidates by exact discriminant bit length, coefficient-height bits, then exact squared L2 norm. For the general-cubic model it orders by coefficient-height bits and exact squared L2 norm. General-cubic Step 8 and its discriminant are not yet implemented, so the machine result explicitly emits `discriminant_evaluated: false`.

Derived. For signed input order `(a0,a1,a3,-a00,-a2,-a4,-a6)`, completing the square gives

```
(2 a0 y + a1 x + a3)^2 = A x^3 + C2 x^2 + C1 x + C0,
A  = 4 a0 a00,
```

```
C2 = 4 a0 a2 + a1^2,
C1 = 4 a0 a4 + 2 a1 a3,
C0 = 4 a0 a6 + a3^2.
```

The implementation evaluates the standard integral cubic discriminant `18 A C2 C1 C0 - 4 C2^3 C0 + C2^2 C1^2 - 4 A C1^3 - 27 A^2 C0^2` exactly. It is 16 times the usual elliptic discriminant in the monic standard case, so it preserves ordering there and also detects singularity in this generalized form when the leading coefficients are nonzero.

The data flow and claim boundary are shown in the labeled editable diagram and its SVG.

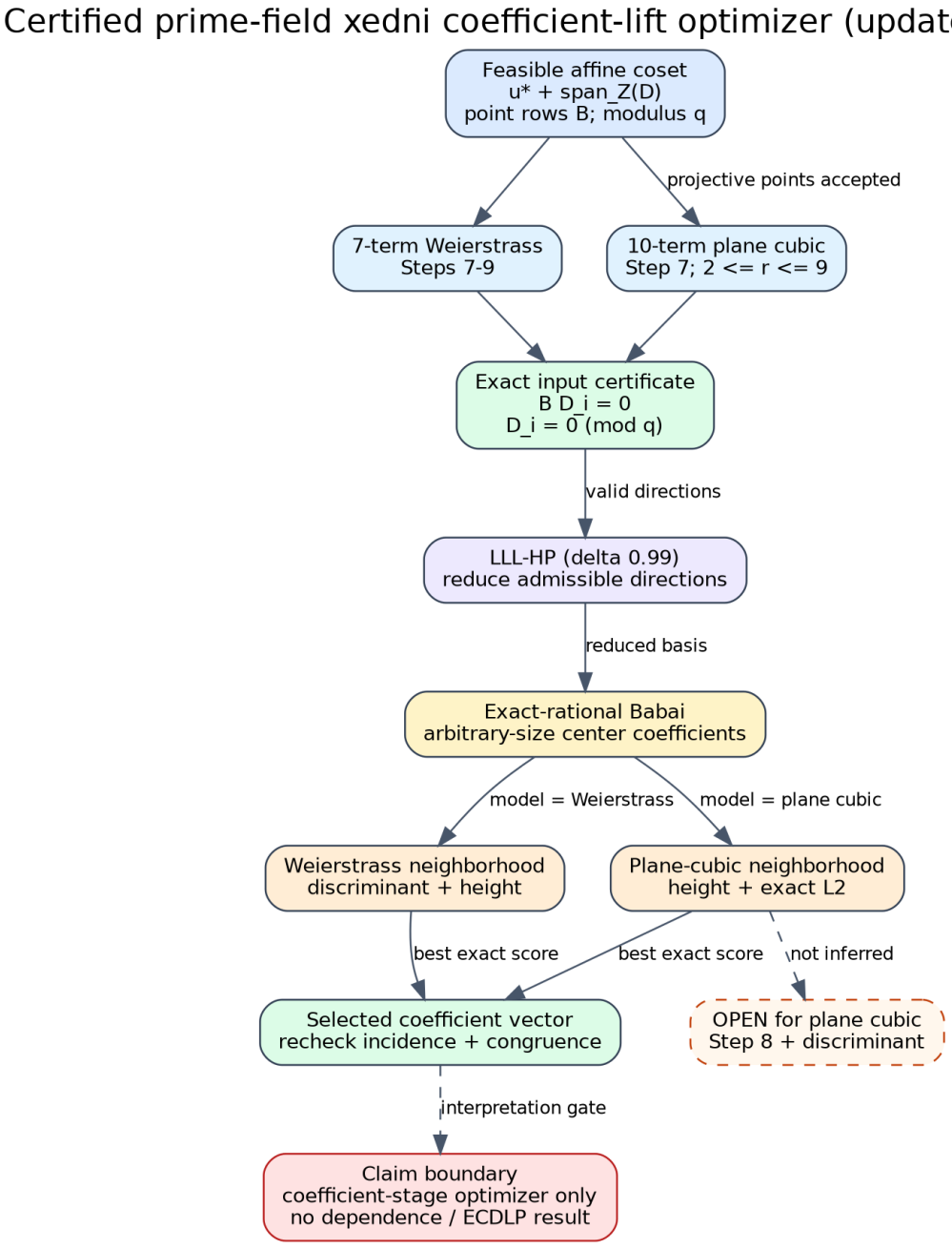


Figure 1: Certified affine-lift pipeline

Verification completed

Verified control. Under the suite's declared Rust 1.87.0 floor, the 2026-11-02 focused validation currently establishes:

- six focused library tests pass;
- both actual `ca-icx` JSON integration paths pass;
- a synthetic rank-three coset containing the known curve y^2+y=x^3-x recovers its signed vector `(1,0,1,-1,0,1,0)` while preserving four incidence rows and modulus 67;
- invalid incidence-changing and noncongruent directions are rejected before search; and
- the discriminant formula returns 1024 for y^2=x^3-x, exactly 16 times its elliptic discriminant 64;
- exact Babai recovers a representative whose center coefficient has magnitude 10^40, removing the former i64 limit; and
- nine rational multiples on y^2=x^3-x+1 produce an exact rank-nine `9 x 10` incidence matrix. General-cubic mode recovers the coefficient vector `(1,0,0,0,0,0,-1,-1,0,1)` while preserving all nine rows and modulus 67.

The complete current-tree release library suite also passes: 3,051 passed, 0 failed, and 90 ignored in 245.29 seconds. Direct touched-file rustfmt checks, `cargo clippy --all-targets -- -D warnings`, and both release CLI integration paths pass on the same tree.

For orientation only, the synthetic input has a 118-bit exact discriminant and 15-bit maximum coefficient, while the planted short representative has a 10-bit exact discriminant and 1-bit maximum coefficient. This is a planted correctness control, not evidence about natural xedni lift quality.

The nine-point fixture is likewise a planted correctness control. Its scalar input representative has 14-bit maximum coefficient; exact Babai and radius zero recover a 1-bit representative. The editable quantitative source, SVG, and source data `nine_point_control.csv` show that exact deterministic comparison. It is not a comparison with HMM-HNF and not a natural-lift result.

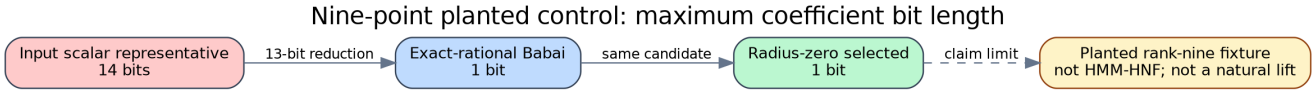


Figure 2: Nine-point planted coefficient-height control

Frozen diagnostic and current evidence boundary

The protocol freezes one JKSST-derived p=67 coefficient coset: curve y^2=x^3+28, generator `(1,30)`, multipliers `(1,5,17,29)`, balanced point lifts, four exact incidence rows, and radii `0,1,2,4`. Its three rows of interest are:

Row	Meaning	Status
raw exact solve	deliberately skew Smith-decomposition representative	input frozen; not a fair HMM baseline
LLL + Babai	modern linear coefficient-height control	implemented but unmeasured on the frozen coset
exact neighborhood	nonlinear discriminant candidate	implemented but unmeasured on the frozen coset

Blocked. No quantitative graph is emitted for the p=67 comparison because there are no run rows to graph. Inventing a plot from planned values or silently substituting the planted control would be misleading. Once the checkpoint gate passes, `run_evaluation.py` produces the raw JSON and the quantitative graph will be generated from those retained rows.

Relation to the asymptotic obstruction and other fields

Source. JKSST Theorem 4.1 conditionally bounds the original prime-field xedni success probability by `C0/p`; their experiments found useful dependent lifts prohibitively rare. Lowering one curve discriminant is therefore a constant-factor engineering result unless a variant explicitly breaks one of the theorem's hypotheses. This optimizer does not claim to do so and does not test point dependence.

Derived. The implementation is prime-field Step-7/9 machinery. It does not construct a lift for `F_{2^m}`. A global number field whose residue field contains general `F_{2^m}` coordinates must have degree at least `m`; Koblitz curve coefficients descending to `F_2` do not make their ECDLP point coordinates descend. The unavailable Kogure 2000/2002 primary texts remain a source gate for a source-faithful characteristic-two variant.

Requirement status

Requested item	Status	Evidence / remaining gap
Reconstruct the lift and "nine point" step	verified complete at linear Step 7	JKSST Section 3.2; ten coefficients, direct point rows, rank-nine control
Implement a modern lift optimizer	verified complete for seven-coefficient mode and general-cubic Step 7	module, exact-rational Babai, two CLI models, exact certificates, focused tests
General-cubic Step 8/discriminant	not implemented	machine output explicitly says the discriminant was not evaluated
Test whether it drastically improves the old algorithm	partial / blocked	frozen p=67 protocol exists; S3 checkpoint gate cannot upload
Prime-field ECDLP improvement	not established	no dependence test or scalar recovery
Binary and Koblitz transfer	not implemented	degree-`m` lift plus Kogure primary-text gate remain

Sources

1. M. J. Jacobson Jr., N. Koblitz, J. H. Silverman, A. Stein, and E. Teske, "Analysis of the xedni calculus attack," *Designs, Codes and Cryptography* 20 (2000), 41–64, DOI 10.1023/A:1008312401197. Primary PDF reviewed, especially Sections 3.2, 4, and 5.4.
2. J. H. Silverman, "The xedni calculus and the elliptic curve discrete logarithm problem," *Designs, Codes and Cryptography* 20 (2000), 5–40, DOI 10.1023/A:1008319518035.
3. G. Havas, B. Majewski, and K. Matthews, "Extended GCD and Hermite normal form algorithms via lattice basis reduction," *Experimental Mathematics* 7 (1998), 125–136.
4. Jun Kogure, "Implementation of the Xedni Calculus algorithm," IEICE Technical Report 100(77), 17–23 (2000), NDL 5409112; metadata only.
5. Jun Kogure, "xedni calculus method (characteristic 2)," ISEC2002-10, 57–60 (2002), NDL 6188214; metadata only.

Reproduction and files

- Code: `suite/src/cryptanalysis/xedni_lift.rs`
- CLI: `ca-icx --json xedni-lift --model weierstrass|plane-cubic --input FILE`
- Nine-point fixture: `suite/tests/fixtures/xedni_lift_nine_points.json`
- Frozen input: `inputs/p67_balanced_snf.json`
- Runner: `run_evaluation.py` (not yet executed)
- Blocker: `problems/xedni-affine-cvp-checkpoint-access.md`
- This PDF: `xedni_lift_optimizer_20261101.pdf`